An Introduction to Ergodicity Economics: Markets (I)

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Markets

Economic agents can maximise their wealth over time by maximising the time-average growth rate of the wealth.

- What does that mean for the portfolio choice?
  - Optimal leverage
- What does that mean for the way markets organise?
  - Leverage efficiency

Geometric Brownian Motion

Geometric Brownian Motion (GBM) is the workhorse model of financial assets:

$$dx = x(\mu\, dt + \sigma\, dW)$$

with drift, μ, and volatility, σ.

Multiplicative dynamics:

Ensemble average growth rate: $g_{\langle\rangle} = \mu$

Time average growth rate: $\bar{g} = \mu - \frac{\sigma^2}{2}$

Portfolio choice

Assume there are only two assets available

- Risk-free asset:

$$dx_r = x_r \, \mu_r \, dt$$

- Risky asset (stock):

$$dx_s = x_s \left(\mu_s \, dt + \sigma_s \, dW \right)$$

Portfolio problem: Allocate wealth to each of the two assets. Leverage, ℓ, is the share of wealth allocated to the risky asset.

Optimal leverage

Portfolio dynamic:

$$dx = x((\mu_r + \ell\,\mu_e)\,dt + \ell\,\sigma_s\,dW\,)$$

where μ_e is the excess drift, $\mu_s - \mu_r$.
The time average growth rate is

$$\bar{g} = \mu_r + \ell\,\mu_e - \ell^2\,\frac{\sigma_s^2}{2}$$

Maximising $\bar{g}$ with respect to ℓ gives us the

Optimal leverage:

$$\ell_{opt} = \frac{\mu_e}{\sigma_s^2}$$

Comparison to standard models

Portfolio choice is usually taught through

- Markowitz portfolio theory:
  Choose a mix between the tangency portfolio and the risk-free asset.
- Sharpe ratio:
  Choose the portfolio with the highest Sharpe ratio: $\frac{\mu_s - \mu_r}{\sigma_s}$

Standard theory recognises the trade-off between risk and return but offers no guidance on how to do the trade-off (optimal choice depends on risk preferences).

Ergodicity Economics does!

Thank you.